Assessing Abdomen

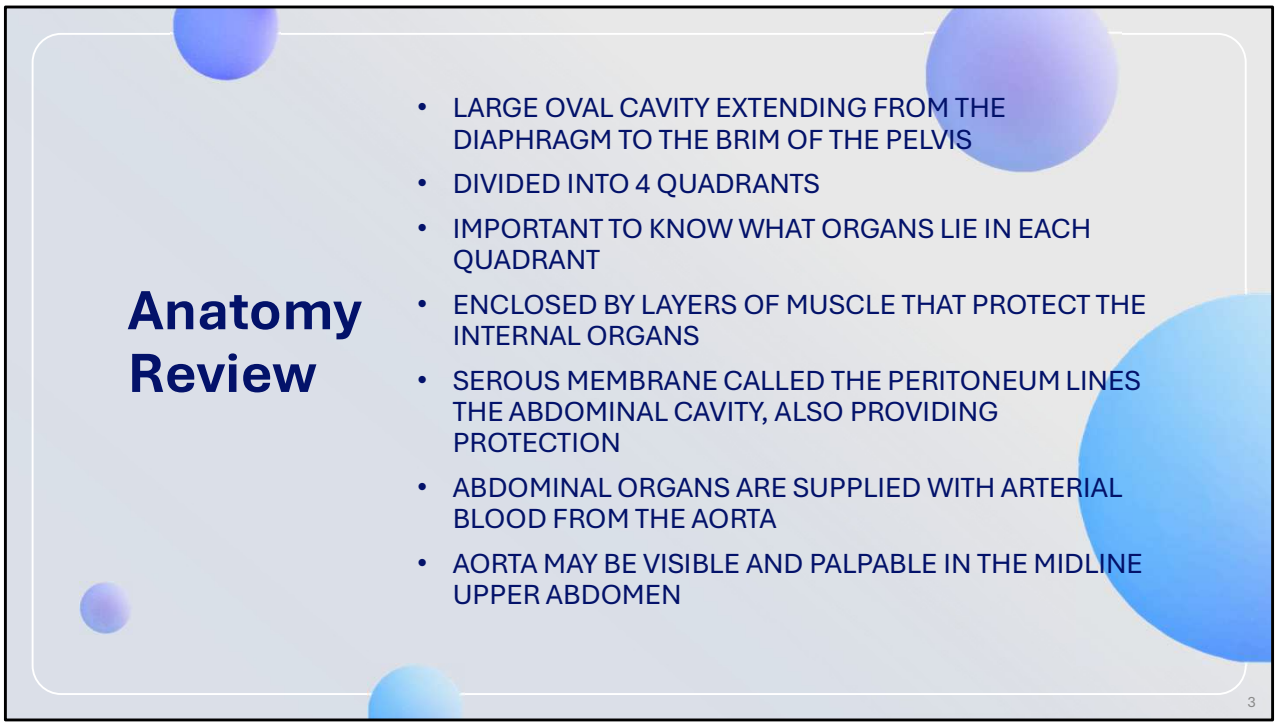
CHAPTER 23

WEBER

NURS3120: HEALTH ASSESSMENT AND PROMOTION

Objectives

1. Describe the structure and function of the abdomen
2. Discuss the risk factors associated with diseases of the abdominal organs and ways to reduce one's risks
3. Differentiate between normal and abnormal findings of the abdomen
4. Analyze the data from the interview and physical assessment of the abdomen to formulate valid clinical judgments



What organs are in the abdomen that we need to consider? Aorta is an artery….
There is an abdomen aortic artery also known as a triple AAA. This runs down the abdomen and breaks off into the femoral arteries. Keep this in mind later during inspection of the abdomen.

Anatomy review continued

- 2 categories of abdominal organs:
- **Solid viscera**: Those organs that maintain their shape consistently
- Liver, pancreas, spleen, adrenal glands, kidneys, ovaries, uterus
- **Hollow viscera**: Structures that change shape depending on their contents
- Stomach, gallbladder, small intestine, colon, bladder

Quadrants of the abdomen

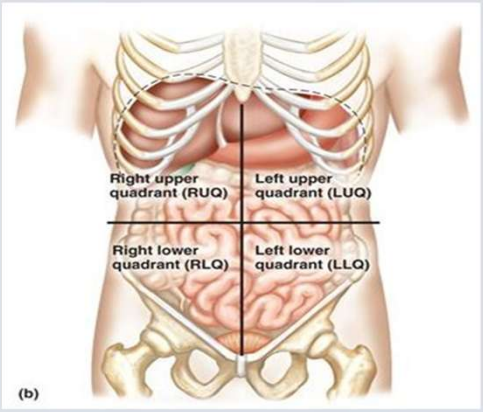
Right Upper Quadrant (RUQ): liver, gallbladder, duodenum, head of the pancreas, right kidney and adrenal, hepatic flexure of colon, part of ascending and transverse colon

Right Lower Quadrant (RLQ): Cecum, appendix, right ovary and tube, right ureter, right spermatic cord

Left Upper Quadrant (LUQ): Stomach, spleen, left lobe of the liver, body of the pancreas, left kidney and ureter, splenic fixture of the colon, part of the transverse and descending colon

Left Lower Quadrant (LLQ): part of descending colon, sigmoid colon, left ovary and tube, left ureter, left spermatic cord

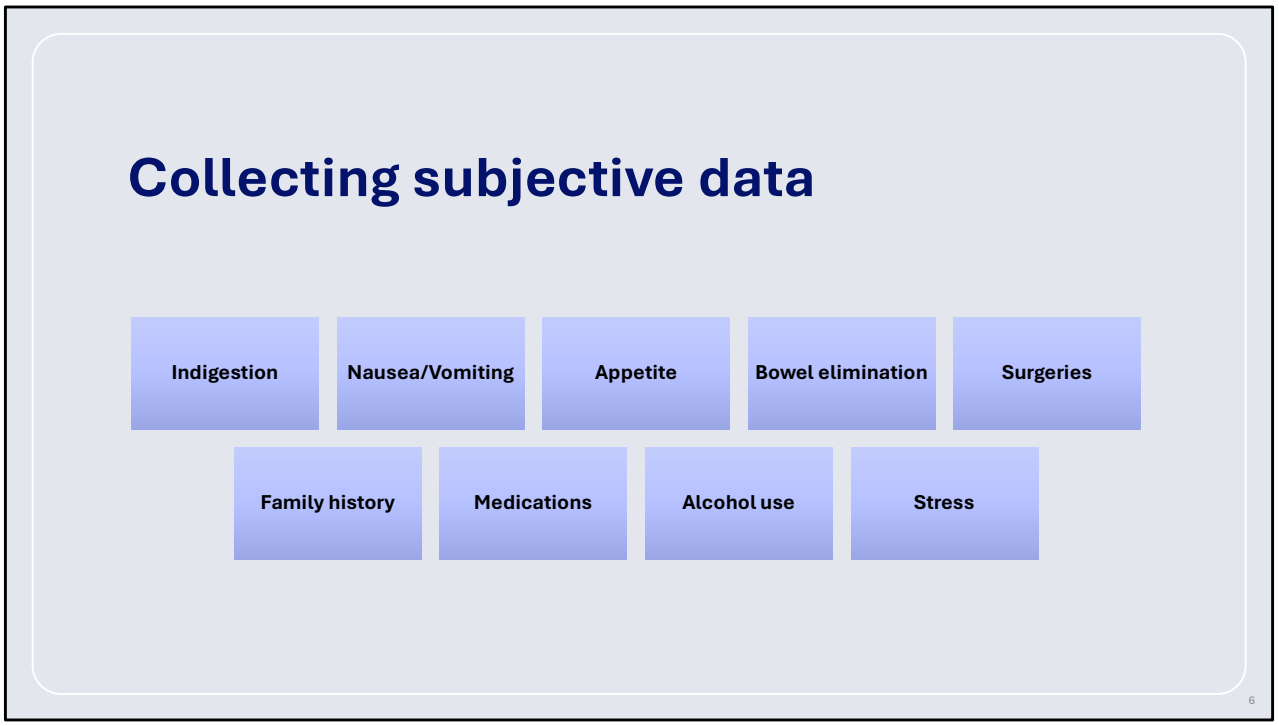
Midline: aorta, uterus (if enlarged), bladder (if distended)



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When you assessing the abdomen either auscultating or palpating please consider the organs and their location in reference to the quadrants of the abdomen.

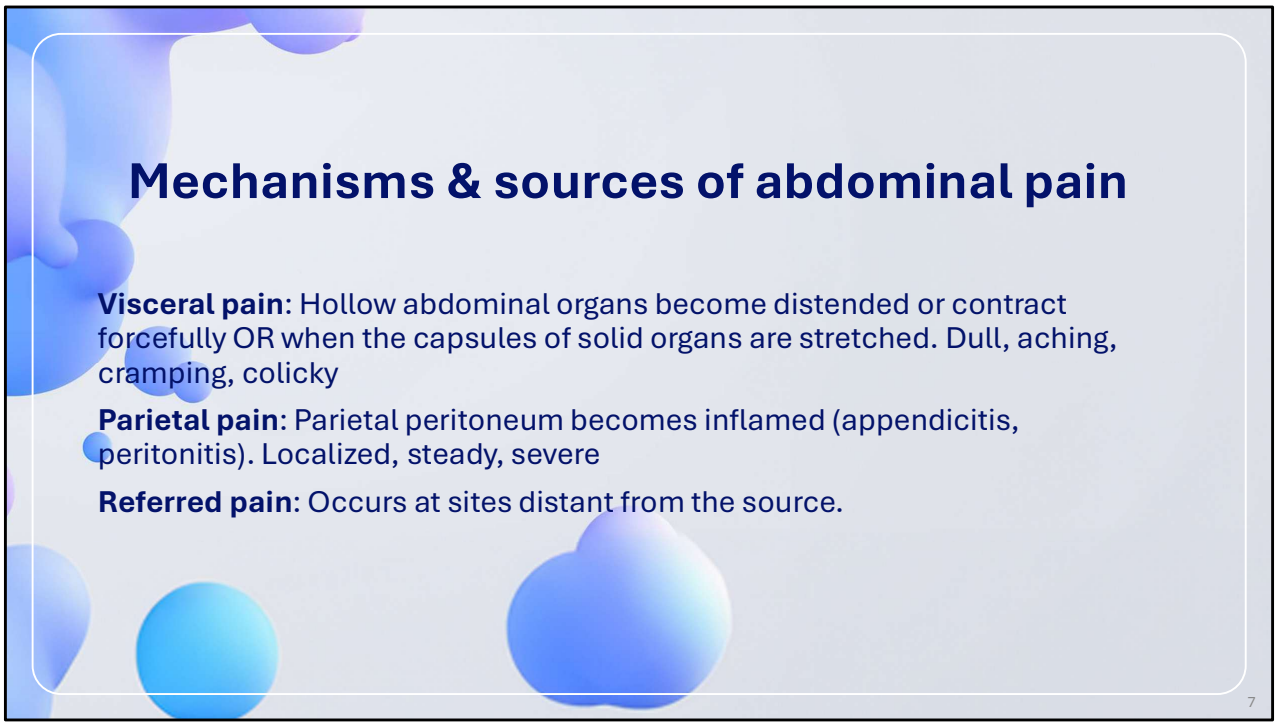
This is the most common method: The abdomen is divided into quadrants, RUQ, RLQ, LUQ and LLQ. We divide vertical down the middle from the tip of the sternum through the umbilicus to the symphysis of the pubis. Then a horizontal line through the umbilicus.



What questions are you going to ask your client?
Indigestion: Heartburn can be an indication of acute or chronic gastric disorders including hyperacidity, GERD, peptic ulcer disease or stomach cancer
N/V: Nausea may reflect gastric dysfunction and is also associated with many digestive disorders and disease of the accessory organs, such as the liver and the pancreas. Nausea may be associated with dietary intolerance, psychological triggers or menstruation.
Why is change of appetite important? Loss of appetite (anorexia) is a general complaint often associated with digestive disorders, chronic symptoms, cancers, and psychological disorders. Significant appetite changes and food intake may adversely affect the client's weight and put the client at additional risks. Older adult clients may experience a decline in appetite from various factors such as altered metabolism, decreased taste sensation, decreased mobility, and possibly depression. If appetite declines, the client's risk for nutritional imbalance increases. Dysphagia: Problems swallowing. Food Intolerance.
Bowel habits: Describe your stools, are they hard or liquid? Color: Why is color important? Changes in bowel habits? Normal varies on the person. Stool characteristics: black stools? Does anybody know what black/tarry stools indicated? Ulcers (can be a slow bleed) or some type of

inflammation of the GI tract: Red or Maroon stools? Active GI bleed in the upper GI tract (esophagus, stomach, duodenum) Is stool hard?

Past surgeries: Previous surgeries can cause abdominal adhesions, thereby predisposing the client to future complications or disorders.
Medications: Does your client need to take a laxative daily?
Acid? Pepto bismol? ASA, Ibuprofen, steroids put people at risk for GI bleeding. Opioids and iron cause constipation
Alcohol: Caffeinated beverages, Cigarette: Long term effects on the organs, stomach, pancreas, and liver. Alcohol related disorders include gastritis, esophageal varices, pancreatitis, and liver cirrhosis. Altered nutrition is another risk for clients who have long term use of alcohol, caffeinated beverages
Stress: Psychological factors can affect GI function through effects on secretion, tone, and motility.



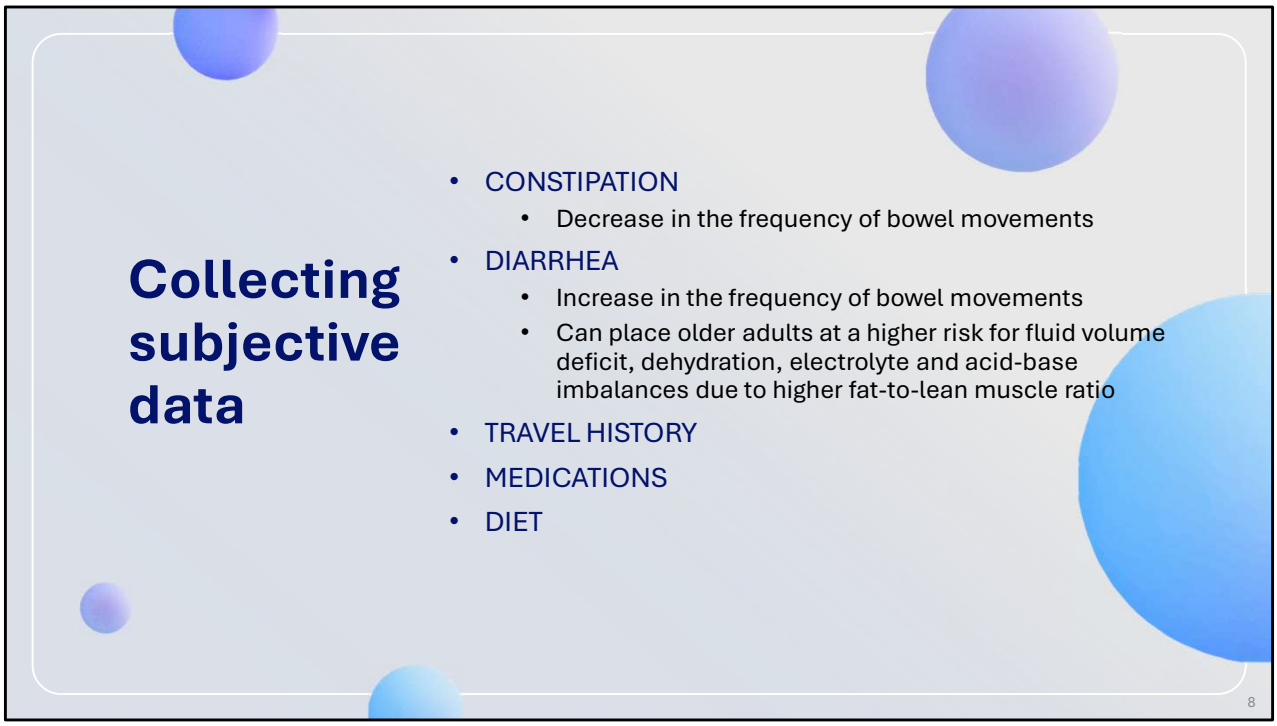
Dull, aching:
Appendicitis
Acute hepatitis
Cholecystitis
IBS
Pancreatitis

Burning, Gnawing:
Dyspepsia
PUD
Colitis
Diverticulitis

Pressure:
Benign prostatic hypertrophy
Prostate cancer
Urinary retention

Sharp, Knifelike:

Splenic rupture
Renal colic
Renal tumor



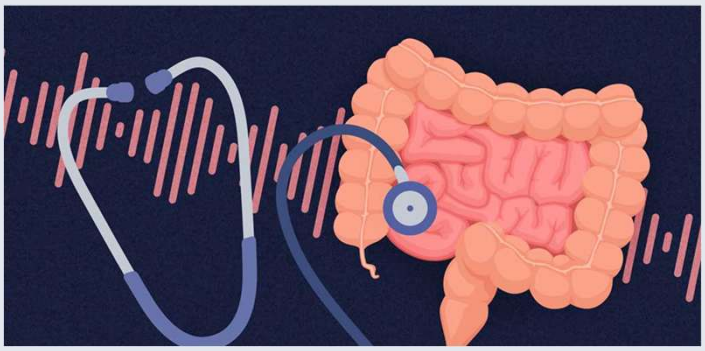
Constipation: In the older client, they will be obsessed with their bowels. Just let them be obsessed. Some people have a stool 2-3 times per day to 3x per week. decreased from your usually bowel routine can be seen as constipation

Consistency: How loose is the diarrhea? Is it watery? Is it brown? bloody or mucoid stools are associated with inflammatory bowel disease (ulcerative colitis and Crohn's disease) Clay colored, fatty stools may be a form malabsorption syndromes.
Older adult clients are at risk for potential complications with diarrhea: **fluid volume deficit, dehydration, and electrolyte and acid base imbalances because they have a higher fat to lean muscle ratio.**
Does it happen right after you eat? Or eat a specific food?
Do you have nausea or vomiting with the diarrhea?
Have you traveled outside of the country?
Medications? Does anybody know what medications cause diarrhea where there is a concern for something else? C-Diff (Clostridium difficile colitis is caused when the normal bacteria in the stomach gets disrupted). This can be fatal problem.
Also you want to know about their medical history…patients

with Crohn's disease or IBS will have diarrhea and you need to know how often they are going to the bathroom and are they on medications to help control their disease.

Collecting objective data

Inspection
Auscultation
Percussion
Palpation



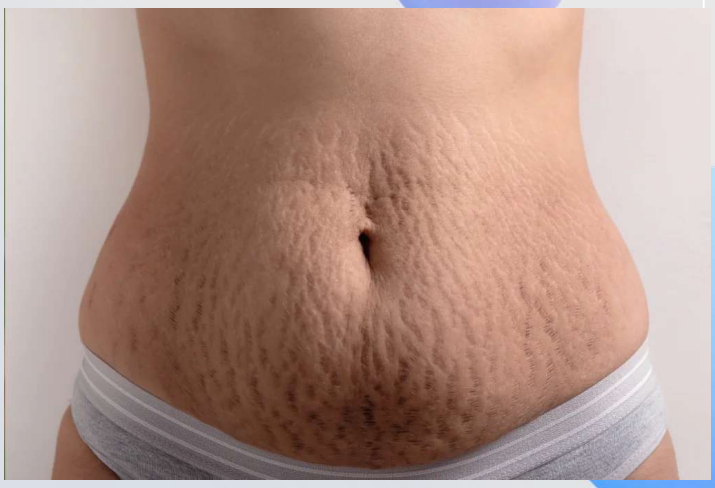
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NURSING CONSIDERATIONS UPON PREPARATION OF THE ABDOMINAL ASSESSMENT:

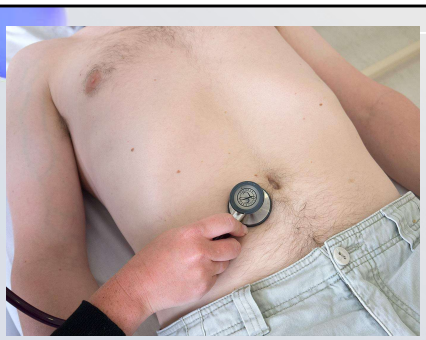
- Pt. empty bladder
- Warm room
- Patient supine, with pillow under head
- Stethoscope warm
- Painful areas last to avoid muscle guarding
- Use distractions to limit abdominal guarding

Inspection

- COLOR
- CONTOUR
- STRIAE
- SCARS
- LESIONS/RASHES
- UMBILICUS
- AORTIC PULSATIONS
- SYMMETRY



Inspect: contour of the stomach – flat, scaphoid (sunken), rounded, protuberant/distended (curves outward):
Abnormal: Distention could be caused by the 6Fs: fat, feces, fetus, fibroids, flatulence and fluid. Tell me what each of the 6F's look like? Fluid AKA Ascites; skin is tight and flanks are also involved
Scaphoid (sunken) abdomen may be seen with severe weight loss or cachexia related to starvation or terminal illness
Is the abdomen symmetric: umbilicus is it midline: Asymmetry may be seen with organ enlargement, large masses, hernia, diastasis recti or bowel obstruction
Skin is it smooth and even in color, observe the color of the skin too.
Are there striae (stretch marks)? Dark blue-pink striae are associated with Cushing syndrome. Normal striae with look pink or blue then turn silvery white.
can you see pulsations from the aorta, hair distribution?
Observe for aortic pulsations: Abnormal would be a vigorous, wide, exaggerated pulsations that may be seen with abdominal aortic aneurysm.





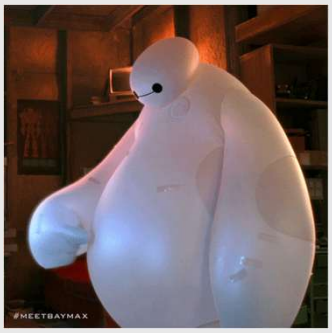
Auscultation

RIGHT LOWER QUADRANT
RIGHT UPPER QUADRANT
LEFT UPPER QUADRANT
LEFT LOWER QUADRANT

Auscultate: Using the diaphragm, begin in the RLQ and proceed clockwise covering all quadrants. bowel sounds occur every 5-15 seconds. Normal is a rate of 5-30 per minute
Hyperactive bowel sounds is abnormal is above 30/minute Hypoactive bowel sounds: below 5 per minute and absent is no bowel sounds. You need to listen for 5 minutes before determining that there are no bowel sounds
"Borborygmus": Loud, prolonged gurgles…"Stomach growling"
Causes of hyperactive BS: early bowel obstruction, gastroenteritis diarrhea,
Causes of hypoactive BS: paralytic ileus, late bowel obstruction, surgery
No bowel sounds: absence of bowel motility could need an immediate referral: paralytic ileus or possible obstruction
vascular sounds – check over the aorta, renal arteries iliac and femoral arteries – usually no sound is present. Listen for a bruit.: They are not normally heard over abdominal aorta.

Postoperatively, bowel sounds resume gradually depending on the type of surgery. The small intestine functions normally in the first few hours postoperatively; stomach emptying takes 24–48 hours to resume; and the colon requires 3–5 days to recover propulsive activity.

Palpation



Perform light and deep palpation over all quadrants for:

Masses

Aorta

Liver

Spleen

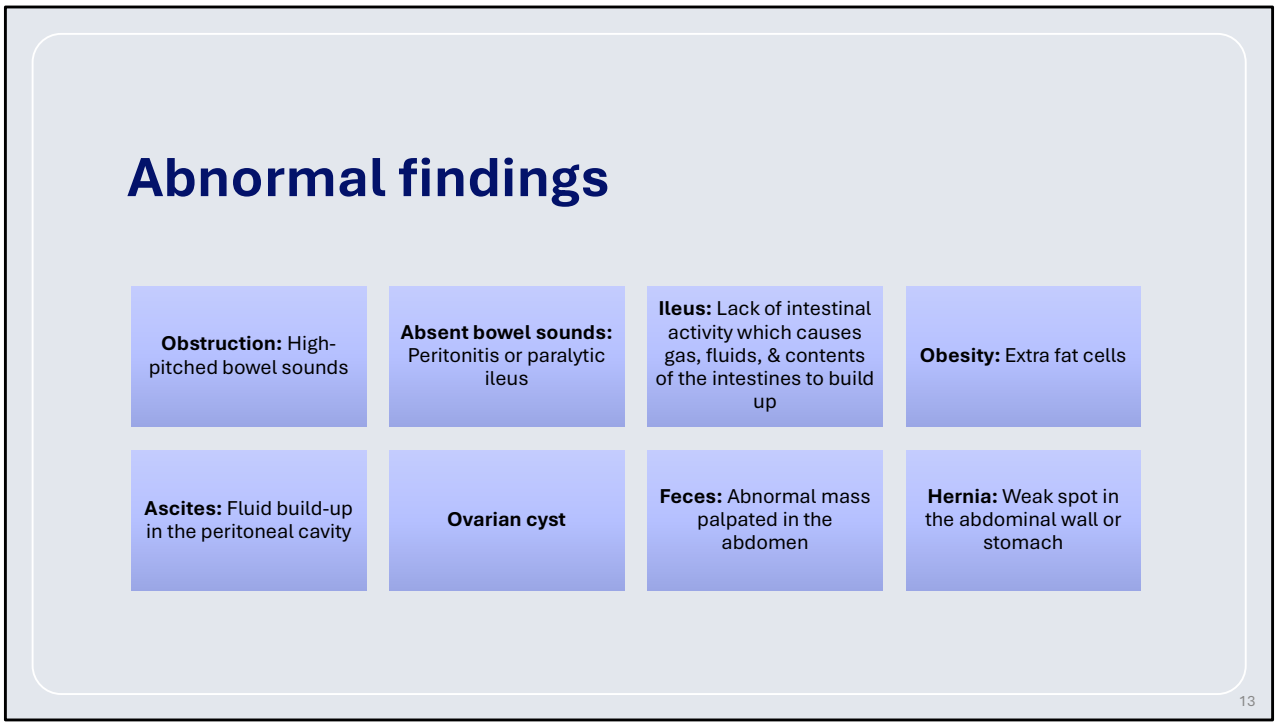
Kidneys

Bladder

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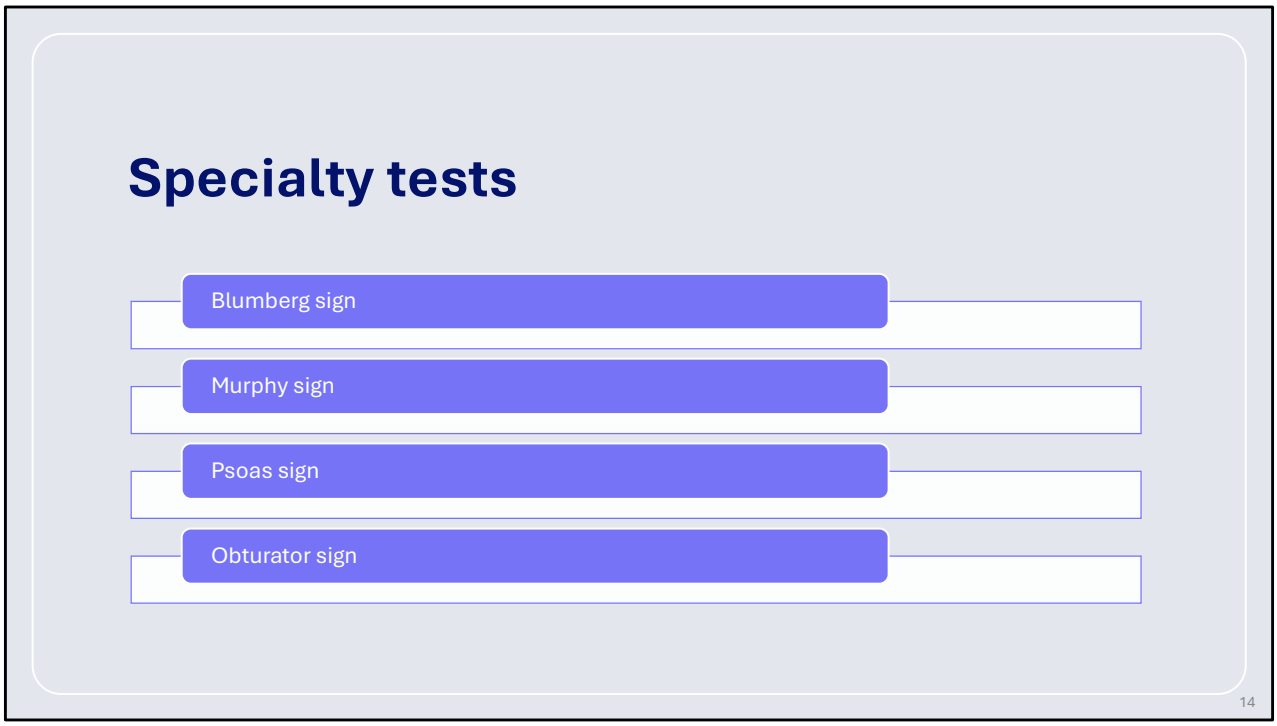
Palpate surface – Light palpation to identify areas of tenderness; 4 fingers in circular motion – forming an overall impression of skin. abd deep areas: Normal is non-tender and soft and no guarding. Abnormal: Involuntary reflex guarding. Severe tenderness or pain may relate to trauma, peritonitis, infection, tumors, or enlarged or diseased organs

Palpate: the liver, spleen (hook Technique is an option), liver under the rt costal margin push up, spleen same but left. – breath in and blow out then push and palpate.
Hook technique: Stand to the right of the client's chest. Curl (hook) the fingers of both hands over the edge of the right costal margin. Ask the client to take a deep breath and gently but firmly pull inward and upward with your fingers.



There is supposed to be air in the stomach which creates a tympany (drum-like percussion).
Air outside of the stomach or intestines is abnormal, on an x-ray it is called free air and is surgical issue
Ascites: fluid build up caused by accumulation of fluid in the peritoneal cavity often related to liver disease

- Obstruction: high pitched bowel sounds
- Absent bowel sounds: peritonitis or paralytic ileus
- Ileus: lack of intestinal activity which causes gas, fluids and the contents of the intestines to build up
- Obesity: extra fat cells
- Ascites: Fluid build-up in the peritoneal cavity
- Ovarian Cyst:
- Feces: abnormal mass palpated in the abdomen
- Hernia: weak spot in the abdominal wall or stomach



Blumberg sign- rebound tenderness. Rebound tenderness is assess by palpating deeply at 90 degrees into the abdomen halfway between the umbilicus and the anterior iliac crest. Abnormal is tenderness is sharp, stabbing pain as the examiner releases pressure from the abdomen…could suggest appendicitis

Murphy sign- Press your fingertips under the liver border at the right costal margin and ask the client to inhale deeply. Normal finding: No increase in pain. Abnormal: accentuated sharp pain that causes the client to hold their breath. pain with inflammation of the gallbladder

psoas sign – Ask the client to lie on the left side. Hyperextend the client's right leg. Pain in the RLQ is associated with irritation of the iliopsoas muscle due to appendicitis.

Obturator test – also used when acute appendicitis is suspected. Right leg is flexed 90 degrees, examiner immobilizes the right ankle with the right hand, left hand rotates the right hip pulling the right knee laterally (external rotation) and medially (internal rotation**).** Pain in the RLQ indicates irritation of the

obturator muscle due to appendicitis or a perforated appendix.

Older adult considerations



- Appearance: Fat deposits in the suprapubic area of women and in the abdominal region “Big Belly” in women
- Salivation
- Esophageal emptying
- Gastric acid secretions
- Gallstones
- Liver size
- Constipation
- Decline in appetite
- Increased risk for dehydration

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Salivation decreases
Esophageal emptying is delayed
Gastric acid secretions decrease
Incidence of gallstones increases with age
Liver size decreases by 25% between age 20-70, liver function remains normal for most
Impaired drug metabolism due to decreased blood flow through the liver
Decline in appetite from various factors such as altered metabolism, decreased taste sensation, decreased mobility and possibly depression
Constipation- reduced stool frequency (less than 3 BM’s per week), straining, lumpy or hard stool, feeling of incomplete evacuation. Risk of constipation is increased by diminished physical activity, decreased fluid intake, decreased fiber in diet, and by ingestion of certain medications, such as iron or narcotics
Impaction – complete blockage by hard, desiccated immovable stool in the rectum, can be life threatening

Older adult clients may experience a decline in appetite from various factors such as altered metabolism, decreased taste sensation, decreased mobility, and, possibly,

depression. If appetite declines, the client's risk for nutritional imbalance increases.

Because muscle mass decreases and fatty tissues increase, the older adult client is at increased risk for dehydration. Sudden weight loss; fever; dry, warm skin; furrowed, swollen, and red tongue; decreased urine output; lethargy; and weakness are all signs of dehydration.

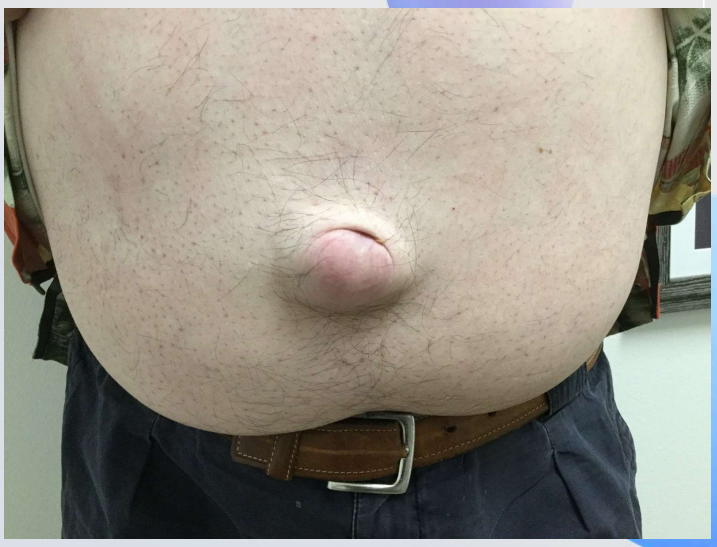
An acute change in mental status (particularly confusion), tachycardia, and hypotension may indicate severe dehydration, which may be precipitated by certain medications such as diuretics, laxatives, tricyclic antidepressants, or lithium.



- Ability to secure food and afford it. May restrict a healthy diet
- May be eating alone, requires more effort than they physically have, may not have the cognitive function to safely prepare meals.
- Ask what they have had to eat in the past 24 hours
- You will soon realize that older adults are OBSESSED with their bowels, and they rightfully need to be because of complications
- What is their bowel routine, what if any laxatives do they take, are there issues with constipation, diarrhea, or fecal incontinence.
- If the client is not eating the correct food due to monetary issues….medication being more important than food…. They might have more issues with their bowel routine

Hernia

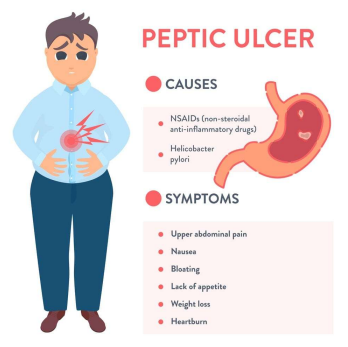
- 4 TYPES:
- INGUINAL
- INCISIONAL
- UMBILICAL
- HIATAL



Can someone tell me what a hernia is?
A weak spot in the intestinal wall and muscle where an organ or fatty tissue squeezes through
There are different types of hernia:
Inguinal hernia: the intestine or bladder protrudes through the abdominal wall or into the inguinal canal in the groin
Incisional hernia: where the abdominal wall pushes through at a site of abdominal surgery
Umbilical hernia: part of the small intestine passes through the abdominal wall near the navel
Hiatal hernia: Upper stomach squeezes through a open in the diaphragm
Diastasis recti: occurs when the bowel protrudes through a separation between the two rectus abdominis muscles. It appears as a midline ridge. The bulge may appear only when the client raises the head or coughs.

Evidence-Based health promotion and disease prevention: Peptic ulcer disease

- Open ulcers/open sores that form in the lining of the esophagus, stomach, or small intestine
- Acid eats away the protective mucous covering & erodes the underlying lining of these organs
- Painful
- Can bleed
- No routine screening available



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Ulcers can be quite painful and can bleed. However, some people experience no pain until the ulcer is quite advanced. If peptic ulcers are left untreated, they can result in internal bleeding, infection, and scar tissue. Symptoms vary widely. Often there is abdominal pain, which can be felt anywhere between the sternum and navel, can cause a burning sensation that often wakes the client in the night, and is worse on an empty stomach (often temporarily relieved with acid-reducing medications or some foods). Ulcers also cause a feeling of fullness that leads to reduced fluid and food intake, hunger, an empty feeling 1 to 3 hours after a meal, or mild nausea. Symptoms may come and go over days or weeks.

Other symptoms may include chest pain, fatigue, weight loss, black or tarry stools, and vomiting, which may be bloody.

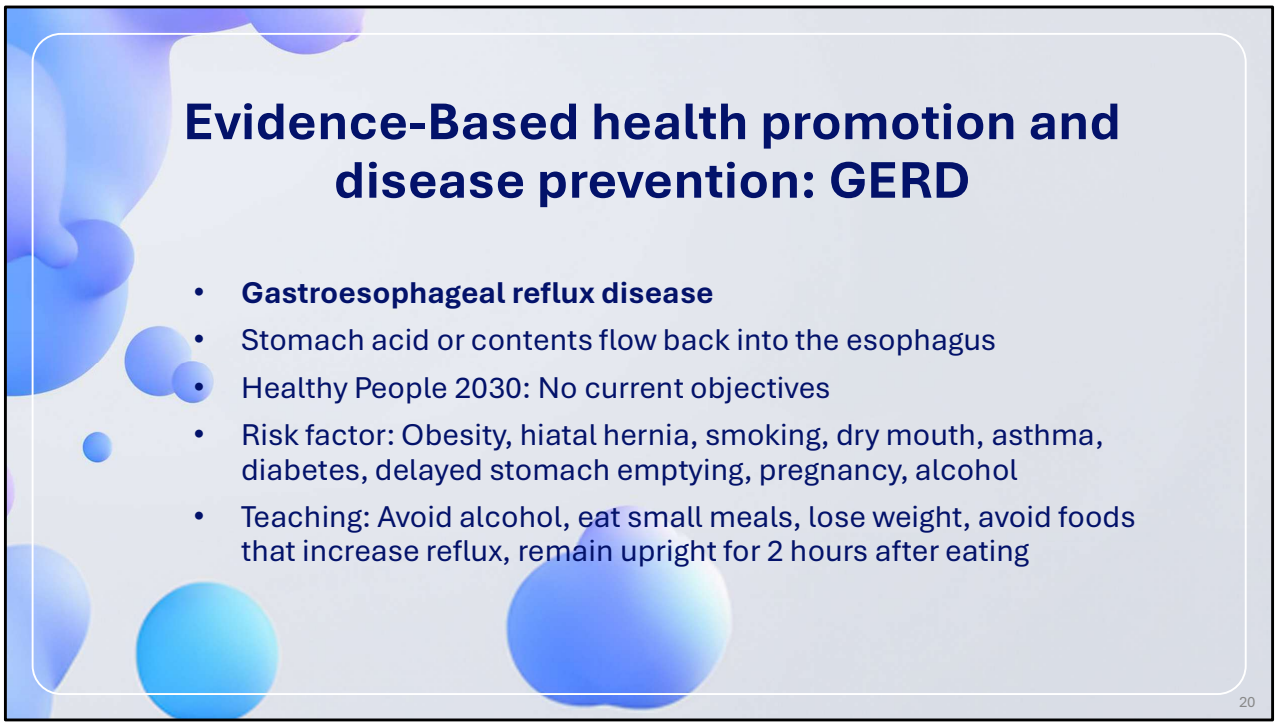
Peptic ulcer disease

Healthy People 2023: No current objectives

Risk factors: NSAID's, smoking or chewing tobacco, stress, spicy foods, alcohol, *H. pylori,* personal or family history

Client education: Hand hygiene, cook food completely, use all recommended precautions when taking medications (take with food), stop smoking or chewing tobacco

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If left untreated, over time, chronic esophageal irritation can lead to serious complications. These complications include narrowing of the esophagus (esophageal stricture), esophageal ulcer, or Barrett esophagus, a condition involving precancerous changes in the esophagus.

Occurs in about 20% of the population
No routine screening

Avoid the foods listed below, which increase reflux:

Food that is very hot or very cold
Fatty or fried foods
Peppermint or spearmint, including flavoring
Coffee, tea, and soft drinks that contain caffeine
Spicy, highly seasoned foods
Tomato-based dishes, such as spaghetti with sauce, chili, and pizza
Citrus fruits and juices, especially in the morning
Chocolate and sweets, if they cause symptoms